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[EXAMPLE] Adaptive wall modelling approach in Large Eddy Simulation with the Immersed Boundary Method.

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Chapter 1

Introduction

1.1 Computational Fluid Dynamics

Computational Fluid Dynamics (CFD) is the study of fluids through a numerical computer-based approach, using algorithms to find solutions to the equations that describe the fluid behavior. The information produced have mainly two purposes: in the field of research, it allows a better understanding of the physical mechanisms governing the fluid motion, and the consequent modelling and control of such mechanisms. In the engineering analyses, fluid simulation is used as a prediction tool, especially during the product design phase, to assess flow characteristics and physical parameters, such as lift and drag forces generated, propagation of acoustic waves or convective heat transfer.

As all mathematical models, the definition of the level of approximation have a primary importance also for fluid simulation. This choice concern various aspects of the description of the phenomena. Firstly the scale of reality considered: the reality can be described at an atomic, particle or micro- and macroscopic level. The last one is that used by classical mechanics, in particular continuum mechanics, that is the case considered in CFD.

The second aspect to be defined in term of level of approximation is the accuracy of dynamic description. It consist on the choice of which forces should be considered exerted on the system, based on their relevance. The most general case is represented by the Navier-Stokes equations complemented by constitutive laws (empirical laws characterizing the fluid and state laws). This case can be simplified, obtaining for example the Euler equations, for the description of inviscid flows.

Another aspect regards the scale separation problem, in particular the spatial and temporal resolution of choice. To obtain a description of maximum quality, all the space-time scales of the solution must be taken into account while solving the unsteady Navier-Stokes

equations. As will be discussed later, it turns out that such a solution criterion is extremely restrictive for multi-scale phenomena, like in the case of turbulent flows.

1.2 Turbulent flows

In most of the engineering applications, flows are prevalently turbulent. Generally this is the case of fluid processed by pumps, compressors or flows around cars, trains, airplanes and many more situations in the everyday surrounding.

Such flows are characterized by some specific properties. A key feature is the multi-scale nature of the vortical structures, whose dimensions span 6 to 8 orders of magnitude. Despite the deterministic nature (they can be described accurately by the Navier-Stokes equations), the strong dependence and sensitivity to initial conditions make their behavior chaotic. Furthermore, turbulent flows are intrinsically three-dimensional and unsteady, they show an high diffusivity (mixing of momentum, energy and species) and also an increased wall friction when compared to laminar flows.

The flow regime is described by the Reynolds number, a non-dimensional value representing the ratio between inertial and viscous forces, and defined as:

$$Re = \frac{\rho\, U_0\, L_0}{\mu} = \frac{U_0\, L_0}{\nu} \tag{1.1}$$

where U_0 and L_0 are the characteristic velocity and length scale of the flow. It is observed that, for higher values of Re, the magnitude of velocity gradients is progressively more intense, and the scale of smaller vortices decreases.

Considering the chaotic nature of the turbulence, a deterministic and instantaneous description of the phenomena is not always the most efficient option. Consequently, the analysis is typically focused on a statistical approach, considering averaged fields. It can be observed how those averaged fields resemble qualitatively the laminar behavior, and their dependence on the Reynolds number diminishes significantly.

1.2.1 Reynolds-Averaged Navier-Stokes equations

To analytically formalize the idea of considering the average fields and the turbulent one separately, the Reynolds averaging operator and the Reynolds decomposition are here introduced. The former consist in a ensemble average over a number of realizations. In more practical terms this implies repeating the same experiment N times, under identical conditions, and averaging the flow variables at the exact same location and time instant across all the independent measures. Considering to apply the Reynolds operator for the

velocity component u:

$$\langle u(\boldsymbol{x},t)\rangle = U(\boldsymbol{x},t) = \frac{1}{N}\sum_{k=1}^{N\gg 1} u_{(k)}(\boldsymbol{x},t) \tag{1.2}$$

Instead for what concerns the Reynolds decomposition, it can be defined as:

$$u(\boldsymbol{x},t) = U(\boldsymbol{x},t) + u'(\boldsymbol{x},t) \tag{1.3}$$

where u' is the fluctuation component and U is the averaged quantity.

By substituting in the Navier-Stokes equations the Reynolds decomposition and applying the Reynolds operator the Reynolds-Averaged Navier-Stokes (RANS) equations are obtained:

$$\begin{cases} \boldsymbol{\nabla}\cdot\boldsymbol{U} = 0 \\ \dfrac{D\boldsymbol{U}}{Dt} = -\dfrac{\boldsymbol{\nabla}P}{\rho} + \boldsymbol{\nabla}\cdot(2\nu\boldsymbol{E}) - \boldsymbol{\nabla}\cdot\langle\boldsymbol{u}'\boldsymbol{u}'\rangle \end{cases} \tag{1.4}$$

where the term $\langle\boldsymbol{u}'\boldsymbol{u}'\rangle$ is the Reynolds stress tensor, and its modelling represents the closure problem of the RANS equations.

1.2.2 Energy cascade

The RANS equations introduced above, allow demonstrating some fundamental mechanisms governing the energy dynamics within turbulent flows. Omitting for brevity the detailed algebraic steps, the mean and turbulent flow kinetic energies can be defined respectively as:

$$\mathcal{K} = \frac{1}{2}\boldsymbol{U}\cdot\boldsymbol{U} = \frac{1}{2}U_i U_i \qquad k = \frac{1}{2}\langle\boldsymbol{u}'\cdot\boldsymbol{u}'\rangle = \frac{1}{2}\langle u_i' u_i'\rangle \tag{1.5}$$

By manipulating the momentum equation, it is possible to derive a formulation for both the time derivative of the mean kinetic energy and turbulent kinetic energy terms:

$$\frac{D\mathcal{K}}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{P}{\rho}U_j + 2\nu\, E_{ij}\, U_i - \langle u_i'u_j'\rangle\, U_i\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_{\mathcal{K}}} \underbrace{-2\nu\, E_{ij}E_{ij}}_{-\mathcal{D}} + \underbrace{\left(\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{-\mathcal{P}} \tag{1.6}$$

$$\frac{Dk}{Dt} = \underbrace{\frac{\partial}{\partial x_j}\left[-\frac{\langle p'u_j'\rangle}{\rho} + 2\nu\,\langle e_{ij}'u_i'\rangle - \frac{\langle u_i'u_i'u_j'\rangle}{2}\right]}_{\boldsymbol{\nabla}\cdot\boldsymbol{F}_k} \underbrace{-2\nu\,\langle e_{ij}'e_{ij}'\rangle}_{-\mathcal{E}} + \underbrace{\left(-\langle u_i'u_j'\rangle\frac{\partial U_i}{\partial x_j}\right)}_{\mathcal{P}} \tag{1.7}$$

where $\boldsymbol{F}_{\mathcal{K}}$ and $\boldsymbol{F}_k$ are the fluxes of kinetic energy (spatial transport terms), $\mathcal{D}$ and $\mathcal{E}$ are the dissipation terms respectively for the mean and turbulent flows and $\mathcal{P}$ is the turbulent

energy production. In a more compact form, the two equation can be written as:

$$\frac{D\mathcal{K}}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_{\mathcal{K}} - \mathcal{D} - \mathcal{P} \qquad \frac{Dk}{Dt} = \boldsymbol{\nabla} \cdot \boldsymbol{F}_k - \mathcal{E} + \mathcal{P} \tag{1.8}$$

Following the approach described by Pope [1], the equations can be integrated over a sufficiently large control volume Ω, surrounding a free shear flow such as a turbulent jet ($Re \gg 1$), in stationary conditions. This perspective, and what result from the integration, allows to identify which component are the dominant in the energy dynamics.

$$\int_\Omega \frac{\partial \mathcal{K}}{\partial t}\, d\Omega = \mathbb{C}_{\mathcal{K}} + \mathbb{F}_{\mathcal{K}} - \mathbb{D}_{\mathcal{K}} - \mathbb{P} \implies \mathbb{C}_{\mathcal{K}} \approx \mathbb{P} \tag{1.9}$$

$$\int_\Omega \frac{\partial k}{\partial t}\, d\Omega = \mathbb{C}_k + \mathbb{F}_k - \mathcal{E} + \mathbb{P} \implies \mathbb{P} \approx \mathcal{E} \tag{1.10}$$

The analysis of the dominant terms in the energy balance equations reveals how the input of energy, associated with convection ($\mathbb{C}_{\mathcal{K}}$) and its viscous dissipation ($\mathcal{E}$) occur at different scales. The apparent decoupling between energy input and output is better described by Richardson's theory.

1.2.3 Richardson's theory

The first concept in Richardson's view of the energy cascade is that the turbulence can be considered to be composed of eddies of different sizes [1].

Considering the eddies with the largest sizes, their characteristic dimensions can be evaluated as: $\ell \approx \ell_0 \approx L_0$ and $u \approx u_0 \approx u'_{rms} = \sqrt{(2/3) \cdot k} \approx (0.1 - 0.3)\, U_0$ (where the last relation comes from observation), thus their Reynolds number:

$$Re_0 = \frac{u_0\, \ell_0}{\nu} \approx \frac{U_0\, L_0}{\nu} \gg 1 \tag{1.11}$$

this suggests how at these scales the direct effect of molecular viscosity is negligible.

Thus, in order to achieve dissipation, unstable large eddies transfer energy to smaller ones. This process continues until the scale $\ell = \eta$ is reached, at this level the associated Reynolds number becomes sufficiently small, or in other words the molecular viscosity begins to be effective in energy dissipation.

The energy transferred to smaller scales can be evaluated as:

$$\frac{k_0}{\tau_0} \approx \frac{u_0^2}{(\ell_0/u_0)} = \frac{u_0^3}{\ell_0} \propto \frac{U_0^3}{L_0} \propto \mathcal{E} \tag{1.12}$$

the scaling of $\mathcal{E}$ with U_0^3/L_0, independently from ν (at the high Reynolds numbers considered) proves to be consistent with the experimental results for free shear flows [1].

1.2.4 Kolmogorov's theory

With the Kolmogorov theory, a better characterization of the involved scales in high-Reynolds flows was proposed. The theory was stated in three hypotheses:

- **Local isotropy hypothesis:** *At sufficiently high Reynolds numbers, the small-scales ($\ell \ll \ell_0$) turbulent motions are statistically isotropic.*

  While larger eddies show an anisotropic behavior, as a result of flow geometry and boundary conditions, scales of motion under a certain length value (small-scales) have statistical properties that are independent of the large, anisotropic scales. Consequently, they exhibit a universal behavior, in the sense that they appear similar in every high-Reynolds number flow.

- **First similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the small-scale motions have a universal form that is uniquely determined by ν and $\mathcal{E}$.*

  As a consequence of this hypothesis, starting from the ν and $\mathcal{E}$ parameters, it is possible to identify a particular set of scales (Kolmogorov scales) with specific length, velocity and time characteristics. This result is obtained by dimensional analysis.

$$\eta = \left(\frac{\nu^3}{\mathcal{E}}\right)^{1/4} \qquad u_\eta = (\nu\mathcal{E})^{1/4} \qquad \tau_\eta = \left(\frac{\nu}{\mathcal{E}}\right)^{1/2} \tag{1.13}$$

  These scales are considered the smallest active scales in the turbulent motion. As a support for this assertion it is possible to observe that the related Reynolds number is unity ($Re_\eta = \eta \cdot u_\eta/\nu = 1$), meaning that viscous forces effectively dominate over inertial forces, allowing the dissipation phenomena.

  It is possible to evaluate the entity of scales separation, considering the ratios of the smallest to largest scales. These ratios depend on the Reynolds number according to the following relations:

$$\frac{\ell_0}{\eta} \sim Re^{3/4} \qquad \frac{u_0}{u_\eta} \sim Re^{1/4} \qquad \frac{\tau_0}{\tau_\eta} \sim Re^{1/2} \tag{1.14}$$

- **Second similarity hypothesis:** *In every turbulent flow at sufficiently high Reynolds number, the statistics of the motions of scale ℓ in the range $\eta \ll \ell \ll \ell_0$ have a universal form that is uniquely determined by $\mathcal{E}$, independent of ν.*

  This intermediate range of scales is known as the inertial subrange. In this region, the eddies are large enough that viscous effects are negligible (since $\ell \gg \eta$), but small enough that the large-scale anisotropic geometry is not felt (since $\ell \ll \ell_0$).

Consequently, the energy cascade is dominated mainly by inertial forces. Again by dimensional analysis:

$$u(\ell) = (\mathcal{E}\,\ell\,)^{1/3} \qquad \tau(\ell\,) = \left(\frac{\ell^{\,2}}{\mathcal{E}}\right)^{1/3} \qquad Re(\ell\,) = \left(\frac{\ell}{\eta}\right)^{4/3} \tag{1.15}$$

In summary, the energy input associated with the mean velocity field (for example as a result of convection) is in part transferred from one point of the domain to another by the flux term, another part is transferred down to the inertial subrange by the production term. This transfer to smaller and smaller scales continues until the Kolmogorov scales are reached, the effects of viscosity become dominant and the energy can be dissipated as heat.

1.3 Simulation techniques

The theoretical framework just introduced, highlights the wide range of scales of motion involved in the turbulent dynamics. In order to describe properly the fluid flow, it is necessary to take into account in the simulation all the space-time scales affecting the phenomena. This can result in an enormous amount of degrees of freedom, especially for high-Reynolds flows.

1.3.1 Direct Numerical Simulation

The most direct approach to perform a simulation, is choosing a space (grid) and time discretization able to represent all the space-time scales, included the smaller ones (Kolmogorov scales), in a non-stationary, three dimensional domain.

In term of accuracy and level of description provided it is unrivalled, however for the number of degrees of freedom involved its cost is extremely high and its applicability is limited to low or moderate Reynolds numbers [1]. These numerical experiments can be considered directly analogous to actual laboratory experiments [2].

The scale separation problem can be illustrated assuming to discretize a domain with dimensions $L \sim \ell_0$. To properly resolve the dissipative structures, the grid spacing Δ must be comparable to the Kolmogorov scale η. Applying the relations (1.14), the number of grid points required along each axis is on the order of $Re^{3/4}$ and in term of time discretization, the required number of time-steps for a τ_0 interval would be on the order of $Re^{1/2}$. Overall, the computational cost of the simulation scales with the Reynolds number, following a $Re^{11/4} \approx Re^3$ relation.

Bibliography

[1] Stephen B. Pope. *Turbulent Flows*. Cambridge University Press, 2000.

[2] J. D. Anderson. *Computational Fluid Dynamics*. Ed. by John F. Wendt. Springer, 1992.